



Multiplicative relationships between terms

Task A: The structure of multiplicative relationships

1) Write the family of rearrangements for the below equations.

a)

$$12 \times 7 = 84$$

b)

$$0.3 \times 2.7 = 0.81$$

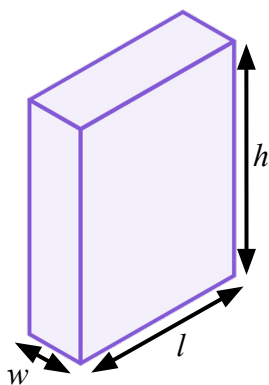
c)

$$7 \times y = 3$$

d)

$$d = s \times t$$

2) The volume of a cuboid can be calculated as 'length $\times$ width $\times$ height'



Fill in the missing parts of these rearrangements.

$$V = lwh$$

V , the **product** of the original multiplicative relationship remained the numerator.

$$\frac{V}{\boxed{}} = lw$$

$$\frac{V}{\boxed{}} = h$$

$$\frac{V}{w} = \boxed{}$$

$$\frac{V}{wh} = \boxed{}$$



Task B: Rearranging multiplicative relationships

1) Write the family of rearrangements for the below equations.

a)

$$\frac{95}{5} = 19$$

b)

$$\frac{0.82}{0.2} = 4.1$$

c)

$$\frac{56}{y} = 7$$

d)

$$\frac{P}{F} = A$$

2) Lucas is rearranging equations.



I just 'swap positions' in the equation.

In each case decide if Lucas is correct and write a sentence to justify your decision.

a)

$$\frac{72}{9} = 8 \text{ can swap to } \frac{72}{8} = 9$$

b)

$$\frac{5}{h} = b \text{ can swap to } \frac{5}{b} = h$$

c)

$$\frac{13}{x} = 2 \text{ can swap to } \frac{2}{x} = 13$$

3) Rearrange each equation into the form $a \times b = c$

a) $\frac{x}{8} = y$

b) $\frac{x}{3y} = y$

c) $\frac{5x}{3yz} = 4z^2$